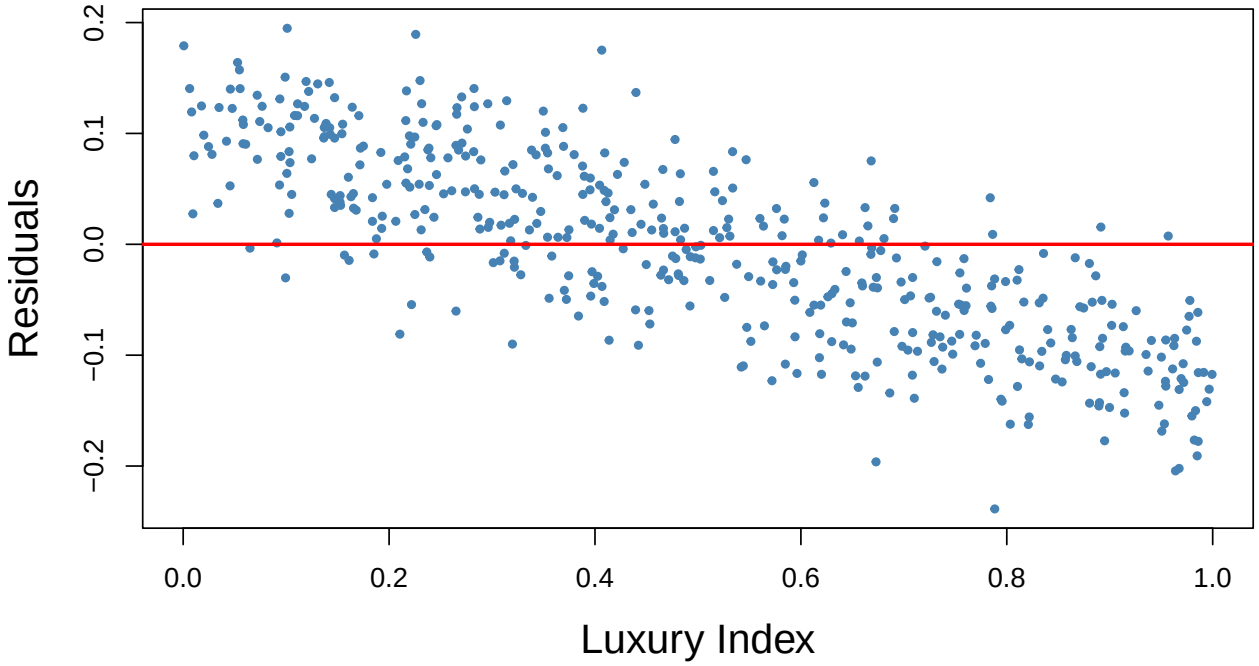
You are interested in the linear relationship between prices and car luxury index. Luxury index is a variable that varies between zero and one.

You started by guessing that $\hat{Y} = 2.25 + 0.70 \times X$, where Y is the logged price and X is the luxury index.

You compute the residuals using this line and plot the residuals against X as below:



The mean of the residuals plotted above is zero.

Now, suppose you would like to refit the line using least squares criterion. Denote the newly fitted line as $\hat{Y} = b_0 + b_1 \times X$. What is the relationship between b_0 and 2.25?

A. $b_0 < 2.25$

B. $b_0 = 2.25$

C. $b_0 > 2.25$

D. There is not enough information